Formula 1 Podium Prediction: Milestone 1 Analytical Report

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1. Task and headline result

We predict, for every driver in a race, whether that driver finishes on the podium (`positionOrder ≤ 3`), using only information available **after qualifying and before the start**. The data is the Kaggle/Ergast Formula 1 dataset (version 24, 1950–2024, 14 relational tables). All numbers and figures in this report come from the run-all notebook `notebooks/f1_podium_prediction.ipynb`, as run on Kaggle.

Grain	one row per (raceId, driverId): 24,586 race starts, 1950–2024
Split	train 1994–2019 · validation 2020–2021 · test 2022–2024 (chronological)
Final model	logistic regression (C = 1) on 20 leakage-safe features
Test (2022–2024)	**PR-AUC 0.726** (primary) · **F1 0.665** (secondary) · ROC-AUC 0.937 · top-3 precision 0.662
Reference rule	"top 3 on the grid finish on the podium": PR-AUC 0.647 · F1 0.649

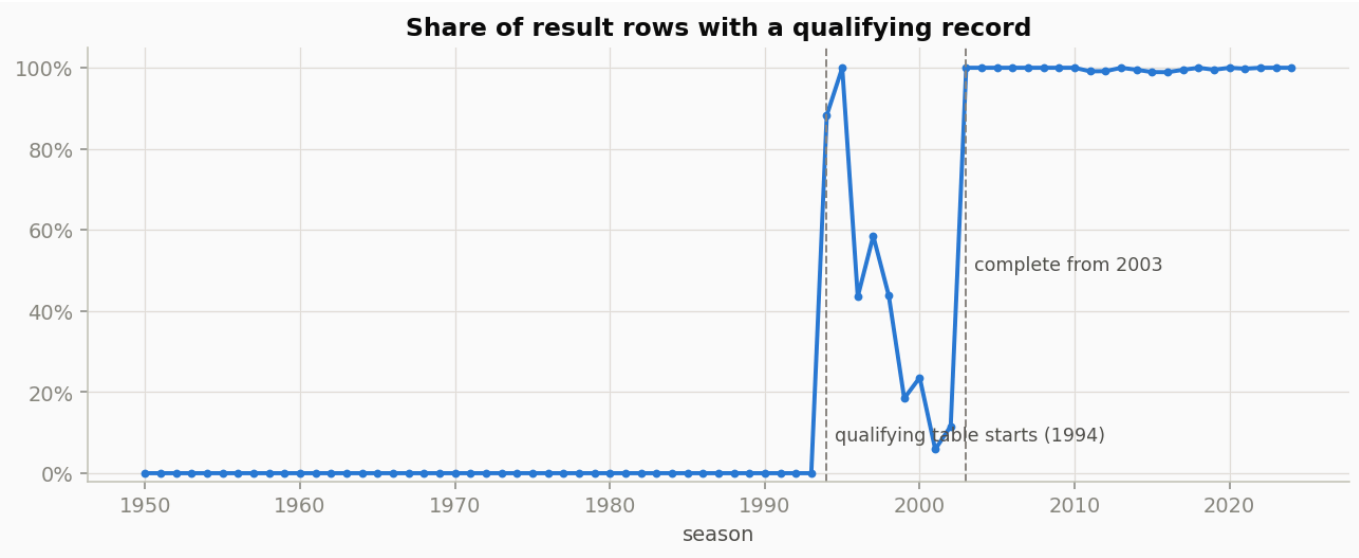
2. Data cleaning and auditing

All cleaning works on copies; the raw CSVs are never modified. Every rule below is backed by a computed audit in the notebook (sections 2–4).

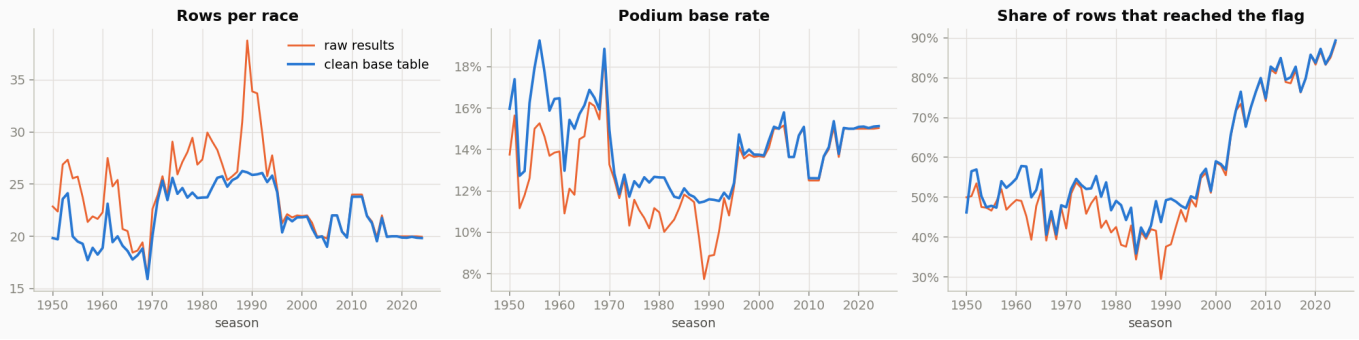
- **Sentinels.**
  - 30 columns in 6 tables contain the literal `\N`, 160,076 cells in total. `q2`/`q3` additionally use *empty* fields for missing values.
  - Both encodings become `NaN`/`NaT`. No `\N` remains after cleaning.
- **Semantic typing.**
  - 12 numeric columns were stored as text, along with 7 date columns, 7 clock times and 7 durations.
  - Dates become datetimes, and race date + time become one UTC timestamp.
  - Durations (`m:ss.sss`) become numeric seconds. The parser was verified against the `milliseconds` columns: the maximum difference is 0.0000 s over 589,081 laps and 11,371 pit stops.
- **Additional issues found.**
  - One impossible qualifying lap: 1995 Japanese GP Q1 = 16:42.640, 10× the pole time. Laps more than 2× the session's fastest time are set to missing.
  - Trailing whitespace in nationalities (`"Argentinian "`).
  - Circuit country written both `USA` and `United States`.
  - 10 qualifying rows with a different constructor than the result row. We use the result's constructor, since that is the car that started.
- **Keys.** All 14 primary keys and 6 natural keys are unique and non-null, and all 27 foreign-key and cross-table checks reconcile. The **only** failure is the modeling grain: `results (raceId, driverId)` has 91 extra rows in

85 pairs.

- **Duplicates (shared drives, 1950–1964 plus one 1978 non-qualifier).** Each duplicate row carries the grid slot of *its car*, so:
  - the “non-zero grid” rule decides **none** of the 85 pairs;
  - the “best positionOrder” rule would give 64 drivers the grid slot of a car they did not start in, which is post-start information.
- **Our duplicate rule.** We keep the driver’s first entry (lowest `resultId`), which is the car the driver started in. We verified this on documented cases: Moss and Brooks in the 1957 British GP, Fangio in the 1956 Italian GP, and Moss in the 1960 Argentine GP. The label is “podium in any car”, because officially these drivers are credited with that result; using only the started car’s position would lose 16 credited podiums.
- **Modeling population.**
  - We remove the **Indianapolis 500**, identified by race name: only 4% of its 107 drivers ever entered another championship race.
  - We remove **non-starters**: `F` (failed to qualify) and `W` (withdrawn).
  - Before 1994, `grid = 0` almost always marked a non-starter. The remaining 87 `grid = 0` rows are pit-lane starts and are placed one slot behind the last car, with a flag.



Qualifying coverage drift: the table starts in 1994, is patchy in 1996–2002, and is complete only from 2003.



Effect of cleaning: the oversized fields and the artificial podium-rate dip of the late 1980s disappear; reliability rose from about 45% to about 88% finishers.

3. Data-engineering questions

Q1. Front-row conversion by circuit, before and after 2014

Joins: `results → races → circuits`, plus `qualifying`.

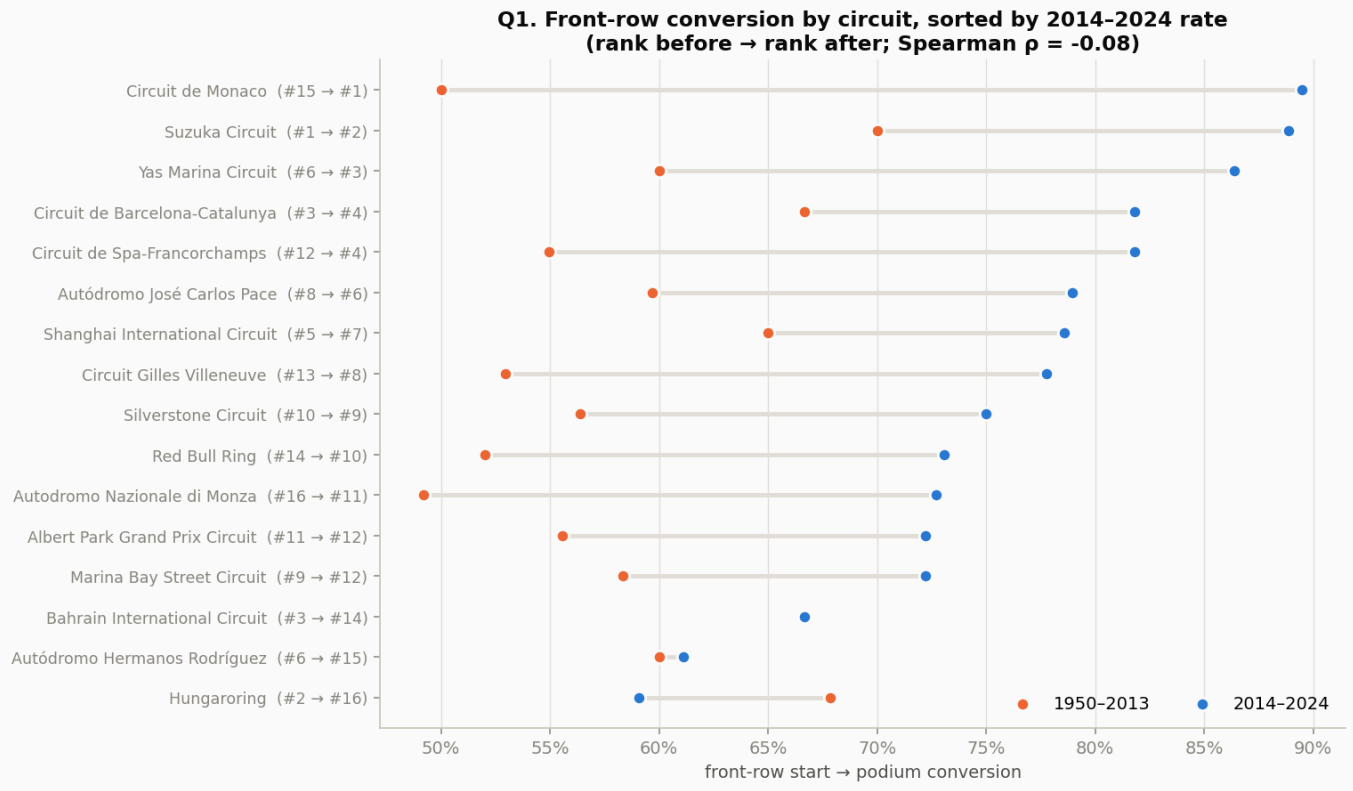
- **Front-row start:** `grid ∈ {1, 2}`, the actual start after penalties.
- **Conversion:** podiums ÷ front-row starts.
- **Ranking:** circuits with at least 10 front-row starts in both periods (16 circuits).

Answer.

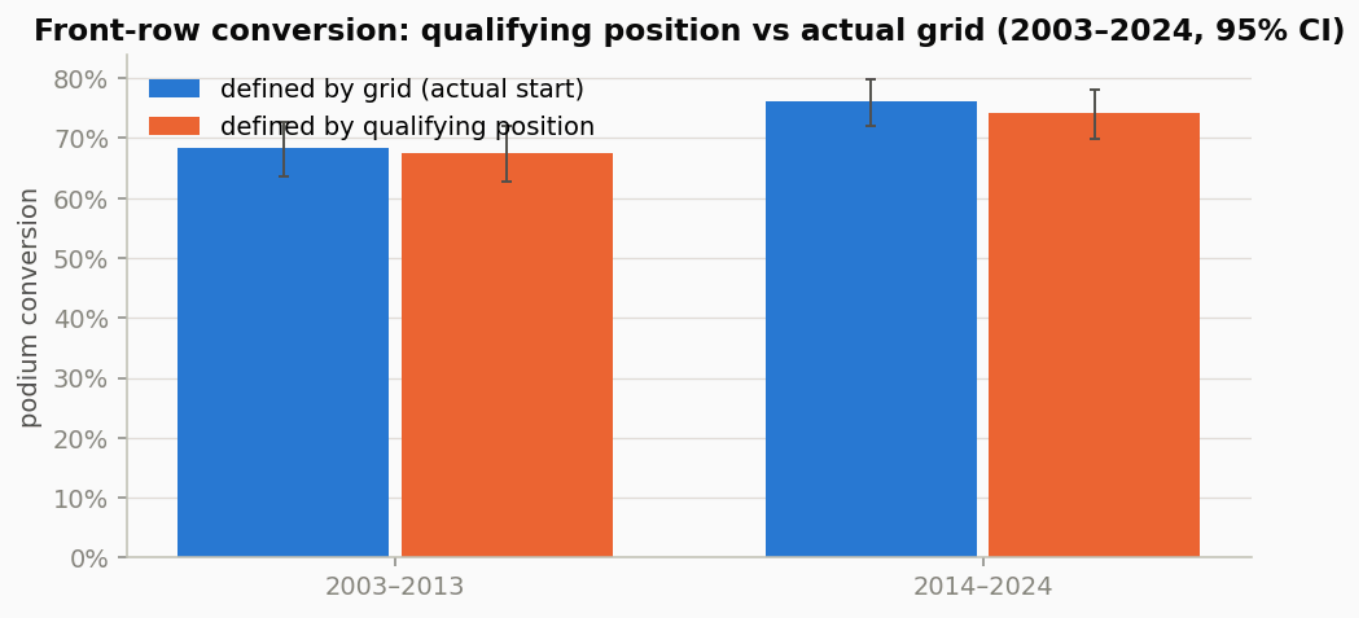
- **A front-row start became far more valuable:** 56.6% conversion in 1950–2013 against **76.2%** in 2014–2024.
- **Best converters in 2014–2024:** Monaco (89%), Suzuka (89%) and Yas Marina (86%).
- **Worst:** the Hungaroring (59%) and Mexico City (61%).
- **The ranking changes almost completely** (Spearman ρ = −0.08, p = 0.76). Monaco went from 15th to 1st; the Hungaroring from 2nd to last.
- **Interpretation:** before 2014 only 70% of front-row starters even reached the flag, against 92% since 2014, so the old ranking was shaped largely by reliability; today it mostly reflects how hard overtaking is. This is plausible but not tested circuit by circuit, and with 14–26 front-row starts per circuit after 2014 only the top and bottom of the ranking are robust.

Qualifying vs grid (2003–2024, full coverage). In 2014–2024, 6.0% of front-row qualifiers did not start on the front row because of penalties, and they converted only 33%, against 77% for drivers who qualified *and* started there. The grid is the right definition of a “front-row start”. The qualifying-based ranking is similar (ρ = 0.90).

Grain: driver-race. **Missingness:** none in `grid`; qualifying is incomplete before 2003, so the qualifying comparison uses 2003+. **Cutoff:** grid and qualifying position are both known before the start.



Q1: front-row conversion by circuit (rank before → after 2014).



Q1: front-row conversion defined by qualifying position vs actual grid.

Q2. Home advantage, controlled for starting position

Joins: `results → drivers` for nationality, and `results → races → circuits` for country. Every nationality is mapped by an asserted demonym-to-country table.

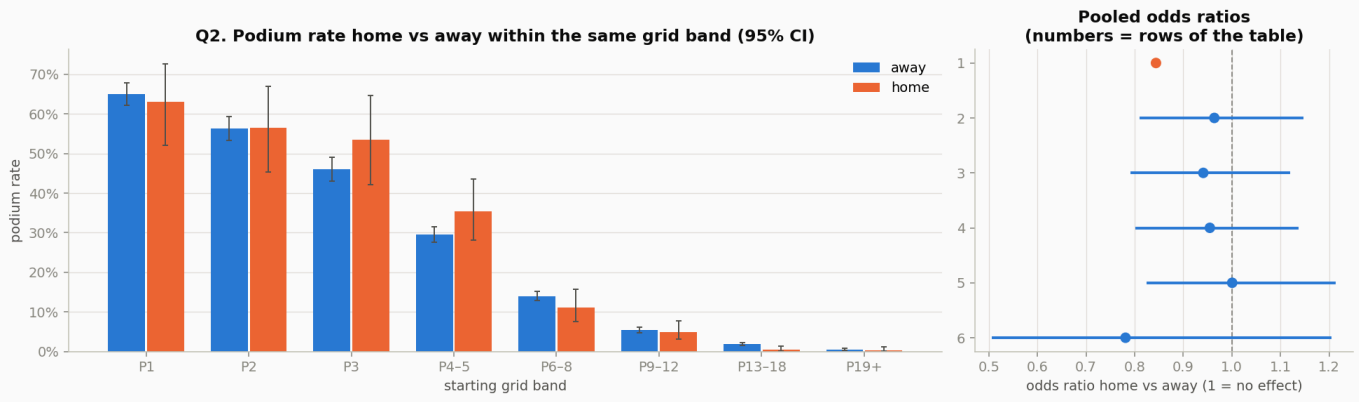
Method: Mantel–Haenszel pooled odds ratio within grid-position bands, with 95% CIs and CMH tests.

Answer: no measurable home advantage once starting position is held fixed.

analysis	odds ratio (95% CI)	p
naive (no control)	0.84	–
within grid band	0.96 (0.81–1.15)	0.71
same driver, same grid band	**1.00 (0.82–1.21)**	0.97
same driver, same grid band, 2003–2024	0.78 (0.51–1.21)	0.32

- **The naive effect is even negative.** Home races attract one-off local entrants who start at the back: 50.6% of home rows start P13 or worse, against 45.5% of away rows.
- **This is an association, not a causal estimate.**

Grain: one row per starter per race. **Missingness:** nationality is never missing and every nationality is mapped. **Cutoff:** nationality and grid are known before the start.



Q2: podium rate home vs away within grid bands, and pooled odds ratios.

Q3. Mechanical-retirement rate by constructor, 2014–2021 vs 2022–2024

Definitions.

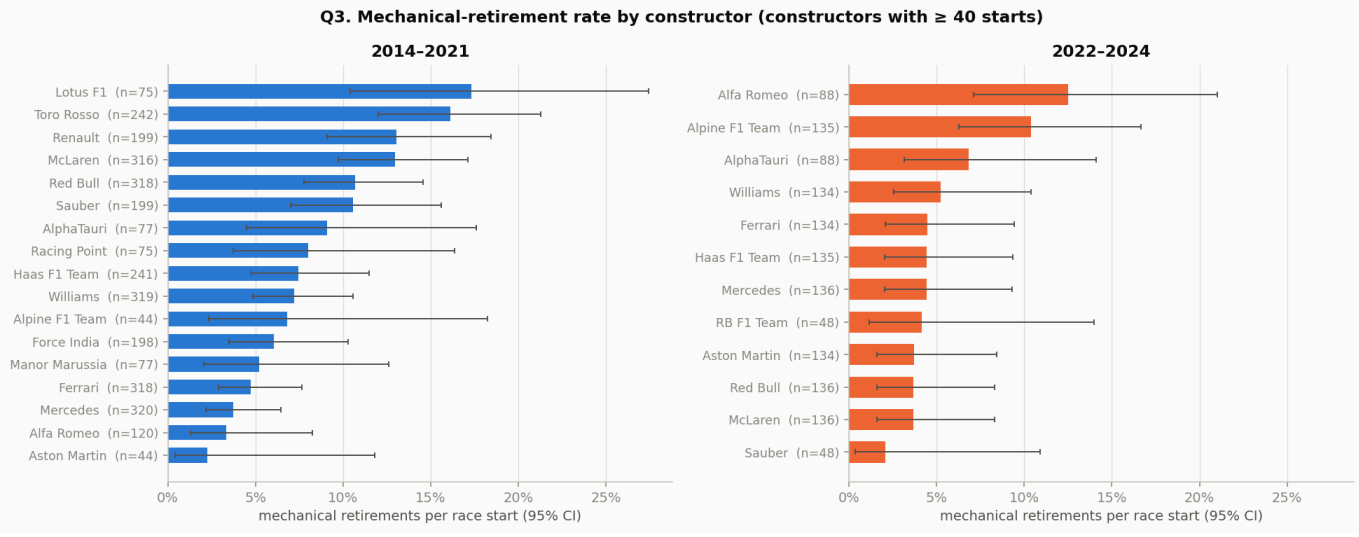
- Denominator: race starts.
- Numerator: starts whose status is a car-component failure.
- All 139 status values are mapped explicitly into finished / mechanical / ambiguous / accident / operational / regulation / health. An assertion fails if any status is left unmapped.
- **Robustness:** a *broad* definition adds the ambiguous statuses (`Retired`, `Wheel`, `Tyre`, …).

Answer.

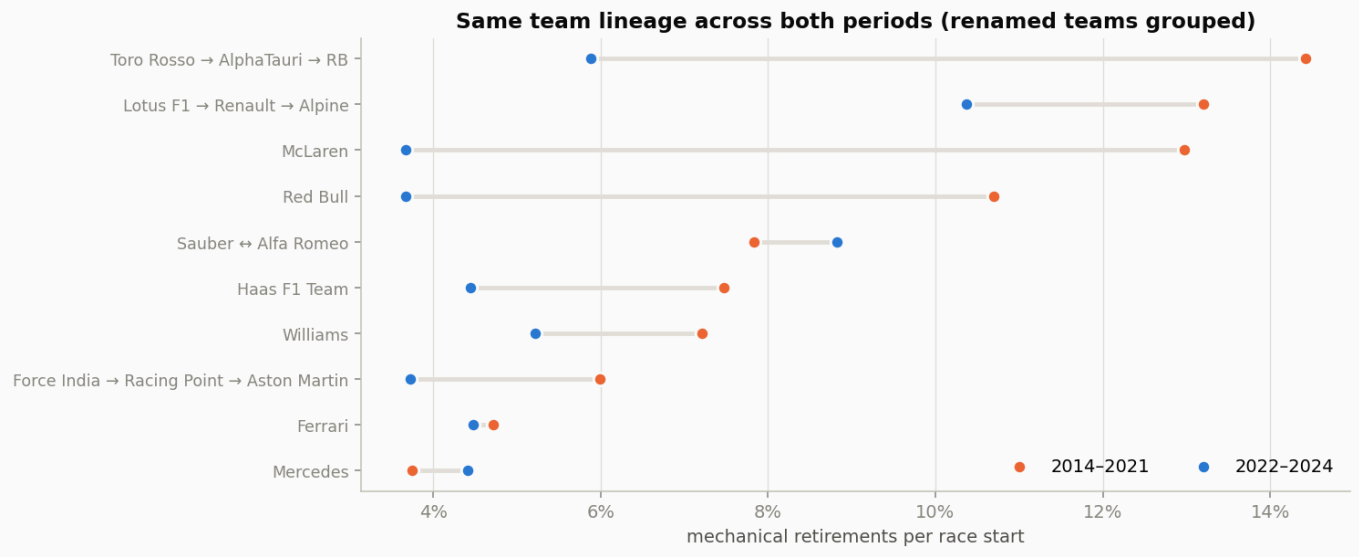
period	highest rates (strict)	field-wide
2014–2021	Lotus F1 17.3%, Toro Rosso 16.1%, Renault 13.1%, McLaren 13.0%	8.9%
2022–2024	Alfa Romeo 12.5%, Alpine 10.4%, AlphaTauri 6.8%	5.5%

- **2014–2021:** the immature Renault and Honda hybrid power units dominate the list. Mercedes (3.8%) and Ferrari (4.7%) were among the most reliable.
- **Strict vs broad:** the ranking is stable for 2014–2021 (Spearman 0.95). It is less stable for 2022–2024 (0.68), where teams sit within overlapping confidence intervals.
- **Renamed teams** are separate `constructorId`s. A lineage view follows each organisation across both periods.

Grain: one row per race start. **Missingness:** status is never missing and every status is mapped. **Cutoff:** status is post-race, so it is used only for this question and for *past-race* reliability features, never as a feature of the current race.



Q3: mechanical retirements per start by constructor (95% Wilson CIs).



Q3: the same team lineage across both periods (renamed teams grouped). Toro Rosso → AlphaTauri → RB improved from 14.4% to 5.9%; Lotus F1 → Renault → Alpine stayed high (13.2% → 10.4%).

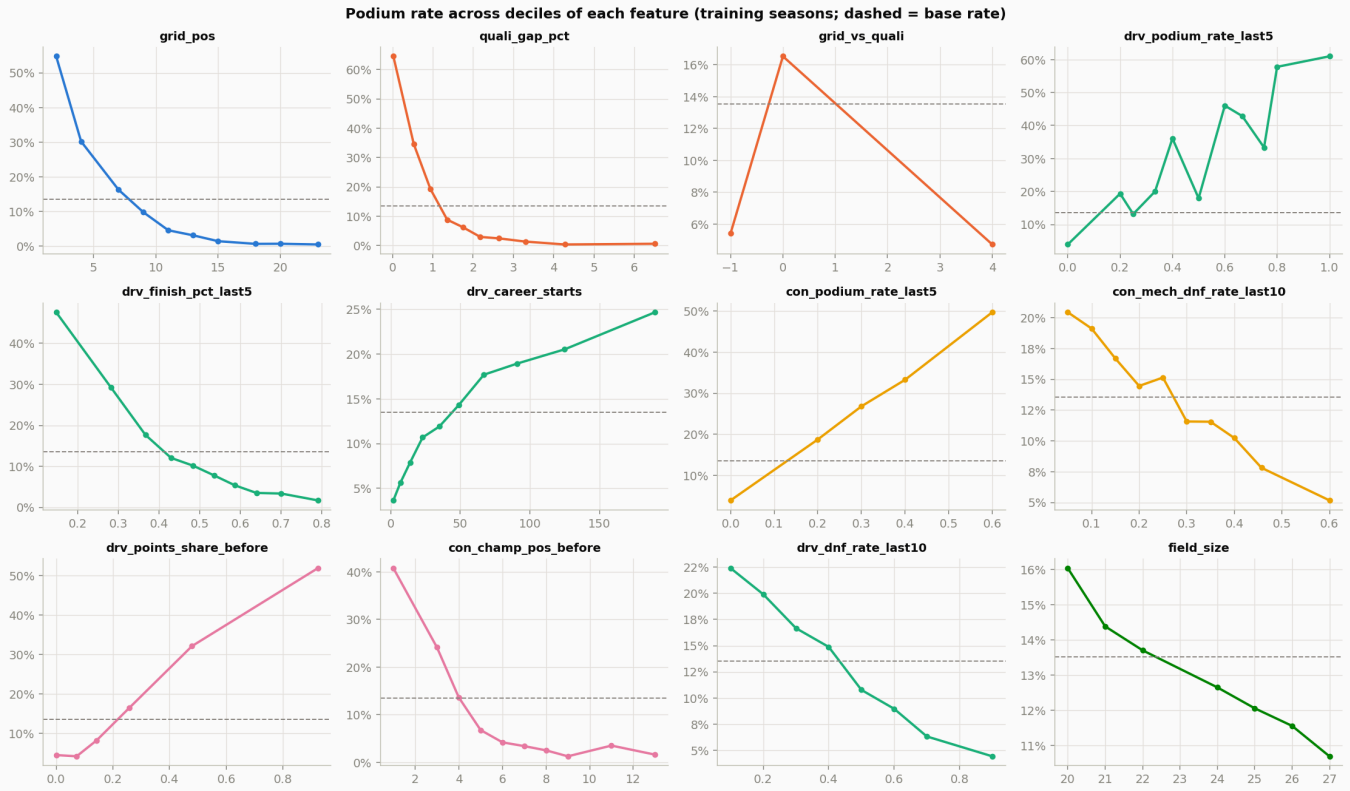
4. Features and why each is valid at the pre-race cutoff

Every *form* feature uses `groupby(...).shift(1)`, so it covers only earlier races. This was verified by a brute-force recomputation on 200 random rows. *Standings* features come from the table after the **previous round** of the same season. The unshifted standings would be leakage: they reach AUC 0.937 against 0.855 for ours, and a perfect 1.000 in round 1.

group	features	why valid before the start
grid	`grid_pos`, `front_row`, `pit_lane_start`	the grid is fixed before the start (penalties applied)
qualifying	`quali_pos`, `quali_gap_pct` (gap to pole), `grid_vs_quali` (penalty size)	qualifying ends before the race
driver form	podium rate and average finish over the last 5 starts, DNF rate over the last 10, career podium rate, career starts	computed only from the driver's previous races
constructor form	team podium rate and average finish over the last 5 races, mechanical-DNF rate over the last 10	computed only from the team's previous races
standings	championship position, points share of the leader and wins, for driver and team, **before** this race	taken from standings after the previous round
context	`field_size`	number of cars on the grid, known once the grid is set

Evidence (training seasons only).

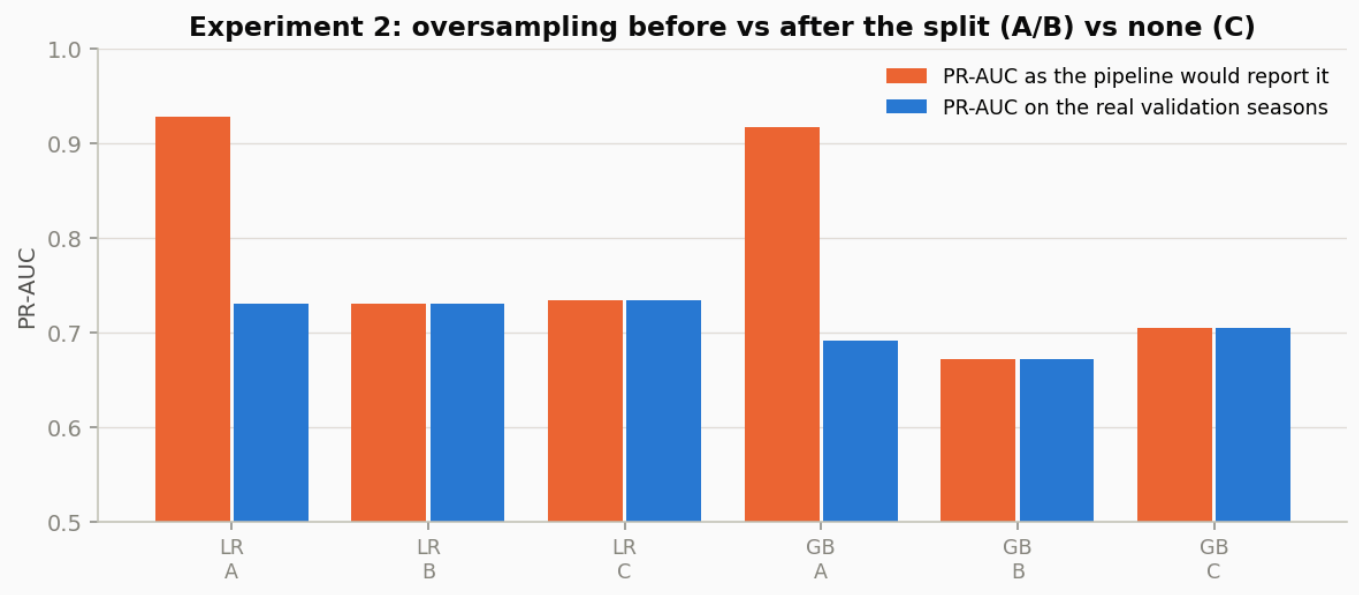
- Univariate AUCs: `quali_pos` 0.91, `grid_pos` 0.88, championship positions 0.83–0.84, constructor form 0.83, driver form 0.79–0.81.
- All of these features are monotone in the decile plots.
- **Excluded on evidence:** `home_race` (Q2, and re-checked: adding it lowers validation PR-AUC by 0.005 and test by 0.001, with a small negative coefficient), sprint results (only from 2021, so absent from training), and `year` (the model would have to extrapolate).



Feature evidence: podium rate across deciles of each feature (training seasons).

5. Split and preprocessing

- **Chronological split:** train ≤ 2019, validation 2020–2021, test 2022–2024.
  - It mirrors real use and keeps races and seasons intact.
  - The test seasons are a genuine regime change (the 2022 rules).
  - All choices (training window, hyperparameters, thresholds, model) use validation only.
- **Training window.** Chosen by experiment from {1950, 1981, 1994, 2003, 2010}. **1994–2019** (the first season of the qualifying table) has the best mean validation PR-AUC, though within 0.002 of 1950 and 1981, i.e. within noise; the short windows (2003, 2010) are worse because they have too little data.
- **Preprocessing:** median imputation with missing-indicator columns, then standardisation. It sits inside the model `Pipeline`, so it is fitted on training rows only.
- **Preprocessing-order experiment 1, fit before vs after the split:** almost no difference here (test PR-AUC 0.7264 for both), even though the means shift by up to 0.17 SD. We still enforce the correct order.
- **Preprocessing-order experiment 2, oversample before vs after the split:**
  - Oversampling *before* the split reports a fake validation PR-AUC of **0.93**; on the real validation races the same model scores 0.73.
  - Even oversampling *after* the split is worse than none (0.731 vs 0.734) and doubles the predicted probabilities.
  - So we do not resample; the threshold is tuned instead.



Experiment 2: oversampling before the split inflates the reported score.

6. Models and comparison

The models below use the same rows, features and preprocessing. Hyperparameters were chosen by validation PR-AUC from 8 logistic-regression, 9 random-forest, 24 gradient-boosting and 12 network configurations. Each model's threshold maximises F1 on validation.

- **Primary metric, PR-AUC:** the positive class is rare (about 15%), and PR-AUC is not inflated by easy negatives.
- **Secondary metric, F1:** the quality of the actual yes/no decision.

Validation (2020–2021):

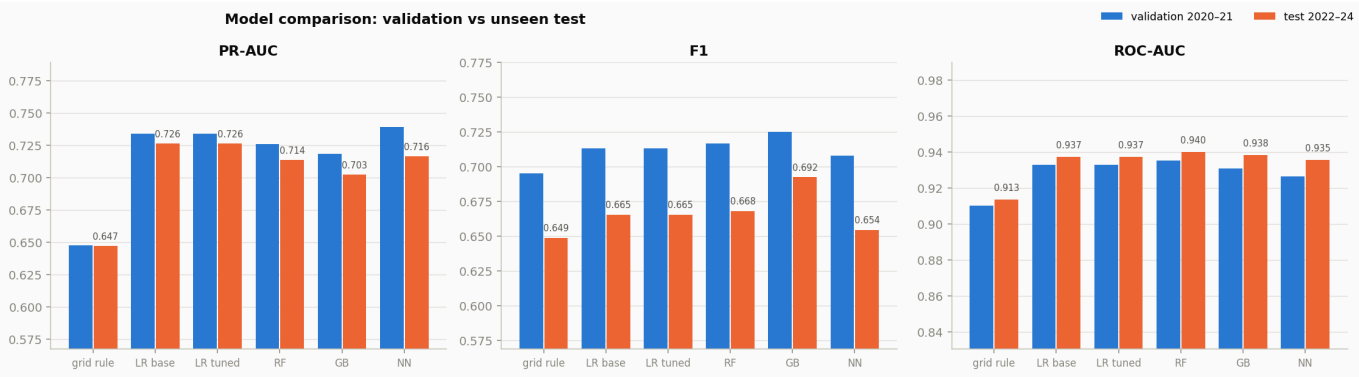
model	PR-AUC	ROC-AUC	F1	precision	recall	top-3 prec.
reference: grid ≤ 3	0.648	0.910	0.695	0.698	0.692	0.701
logistic regression (baseline = tuned, final)	**0.734**	0.933	0.713	0.652	0.786	0.692
random forest	0.726	0.935	0.716	0.636	0.821	0.701
gradient boosting	0.718	0.931	0.725	0.707	0.744	0.675
shallow NN (MLP 64→32)	0.739*	0.927	0.708	0.683	0.735	0.701

Test (2022–2024), plus train PR-AUC:

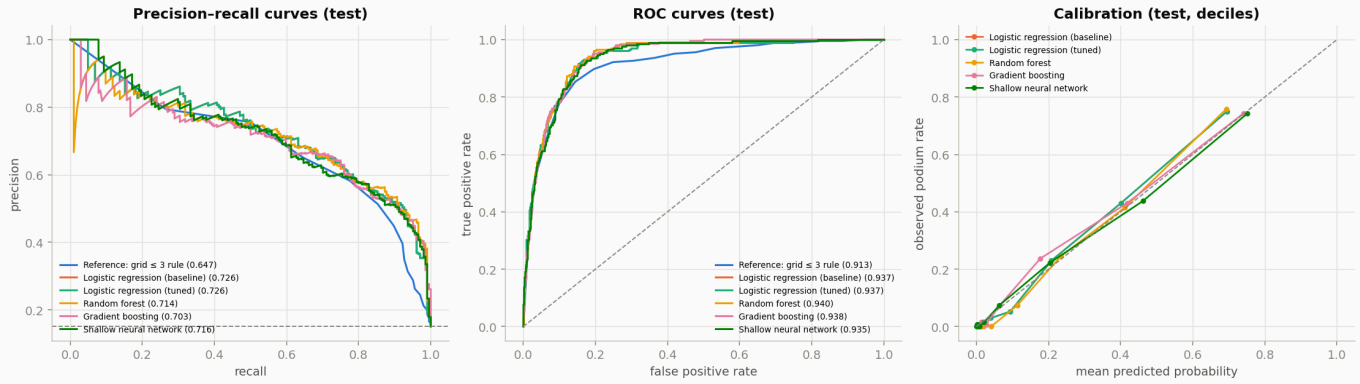
model	train PR-AUC	PR-AUC	ROC-AUC	F1	precision	recall	top-3 prec.
reference: grid ≤ 3	0.590	0.647	0.913	0.649	0.650	0.647	0.652
logistic regression (baseline = tuned, final)	0.655	**0.726**	**0.937**	**0.665**	0.542	0.863	0.662
random forest	0.748	0.714	0.940	0.668	0.537	0.882	0.657
gradient boosting	0.787	0.703	0.938	0.692	0.643	0.750	0.637
shallow NN (MLP 64→32)	0.686	0.716	0.935	0.654	0.614	0.701	0.637

* single seed; the 5-seed mean is 0.727 ± 0.006. The network was therefore selected on its mean, and logistic regression became the final model. Tuning chose C = 1, the default, so the tuned and baseline logistic regressions are the same model. Train PR-AUC is *lower* than validation for logistic regression and the network because the 1994–2002 training seasons (patchy qualifying, low reliability, bigger fields) are harder to predict than the modern seasons; the tree ensembles fit their training rows more closely and score higher there.

- **Learned models vs the rule.** A race-level bootstrap gives 95% intervals of about ±0.06 on test PR-AUC. **All learned models beat the rule** (0 of 1,000 bootstrap samples favour the rule) but are **indistinguishable from each other**: the flexible models beat the baseline in only 4–17% of samples.
- **The models are well calibrated.**
- **Top-3 precision is no better than the grid rule** (0.64–0.66 vs 0.65): the models' advantage is in probabilities and ranking across the whole field, not in picking the top 3.
- **Best and worst test seasons:** 2023 is the easiest (PR-AUC 0.790, Red Bull's dominant year) and 2024 the hardest (0.665).



Validation vs test scores for all models.

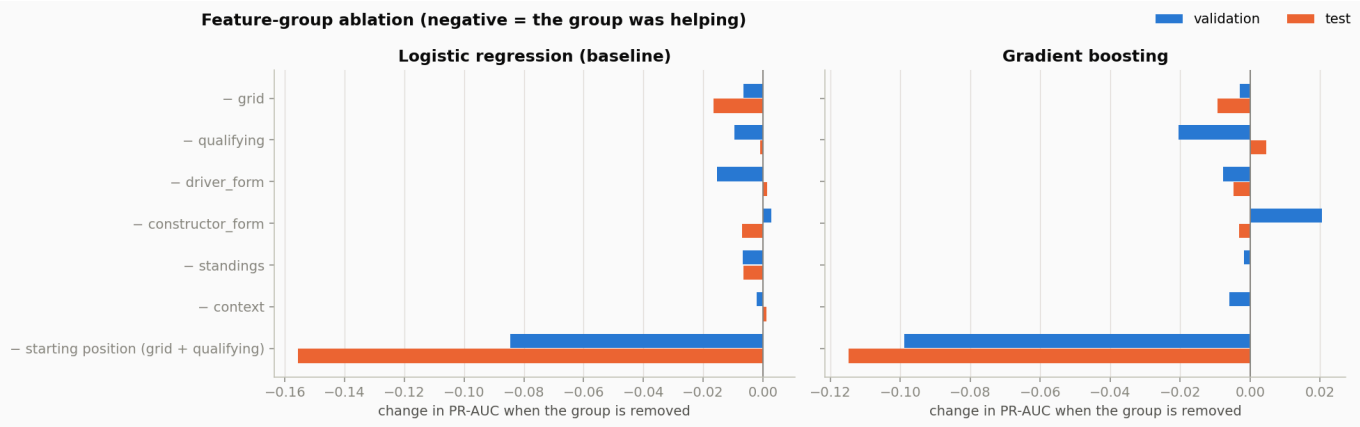


Test precision–recall, ROC and calibration curves.

7. Feature-group ablation

We removed one group at a time and retrained the final model, with gradient boosting as a second opinion.

- **Starting position (grid + qualifying together) is the backbone:** removing it costs −0.16 test PR-AUC for logistic regression and −0.11 for gradient boosting.
- **Removing grid or qualifying alone** costs only 0 to −0.02, because the two are 98% correlated.
- **Driver form, constructor form and standings each add a little** (up to about −0.02, with signs that differ between validation and test), and they substitute for one another because all of them encode “fast car this season”.
- `field_size` **adds nothing** for either model ($|\Delta| \le 0.006$).



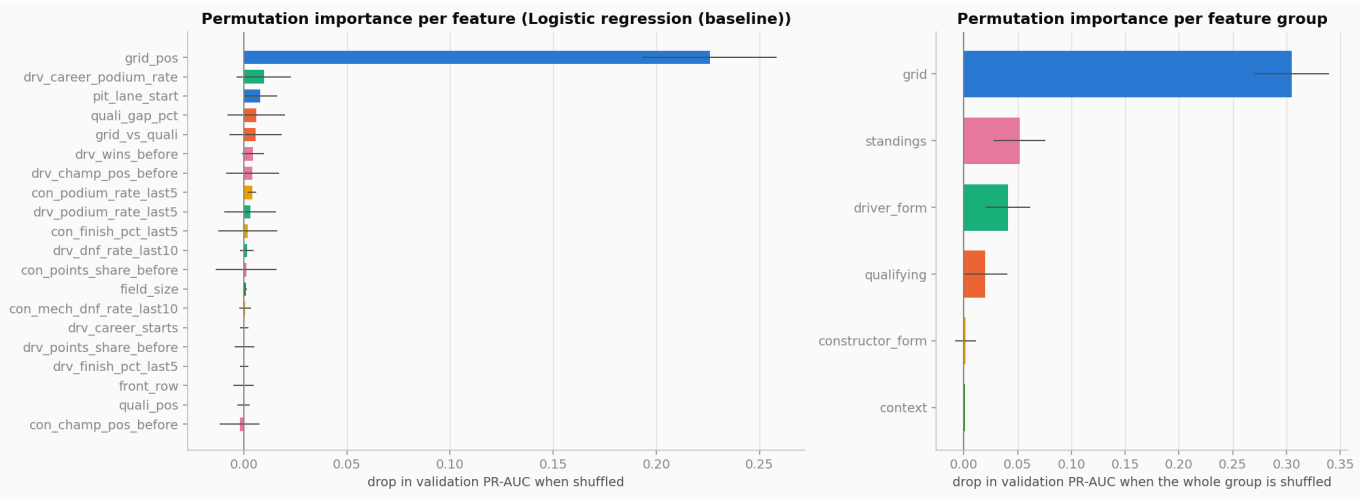
Feature-group ablation.

8. Explainability

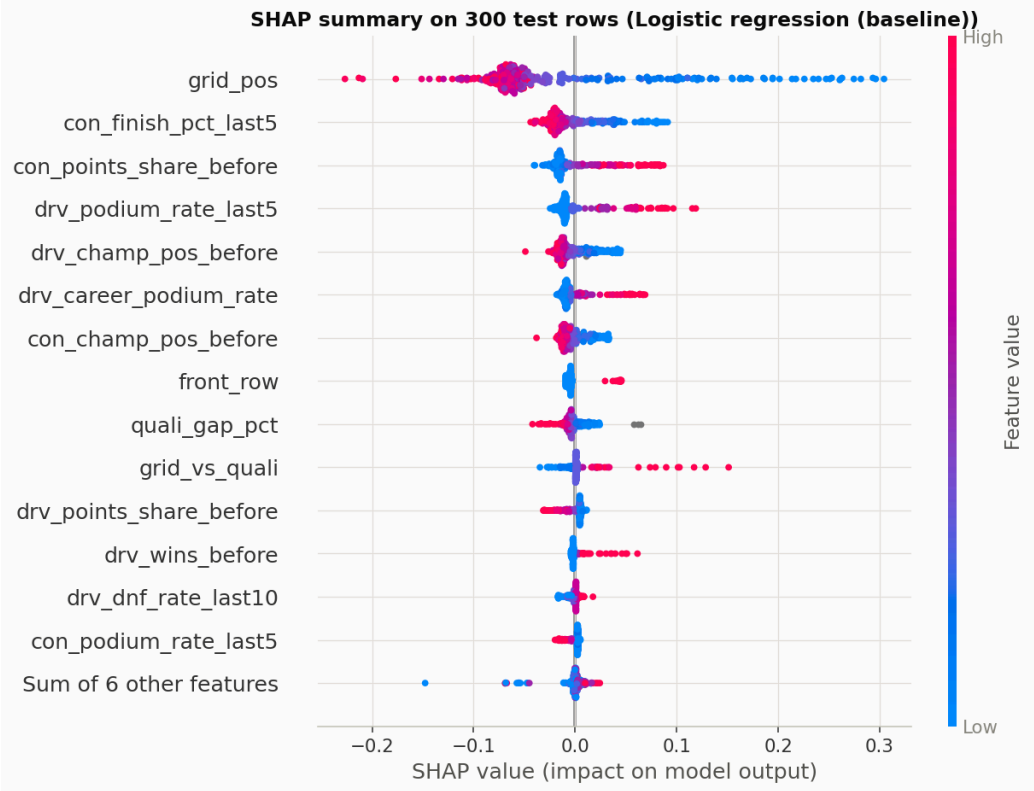
> **Importance is not causation.** These methods show what the *model relies on* to predict, not what *causes* podiums. For example, team form is important because fast cars stay fast, not because past podiums make a car faster. Correlated features also share and swap importance.

- **Global, permutation importance (validation).**
  - `grid_pos` dominates: shuffling it costs 0.23 PR-AUC.
  - Grouped: grid 0.305 ≫ standings 0.052 > driver form 0.041 > qualifying 0.020 > constructor form 0.002 > context ≈ 0.
- **Global, SHAP** (permutation explainer on the full pipeline, background from 2003–2019 rows).

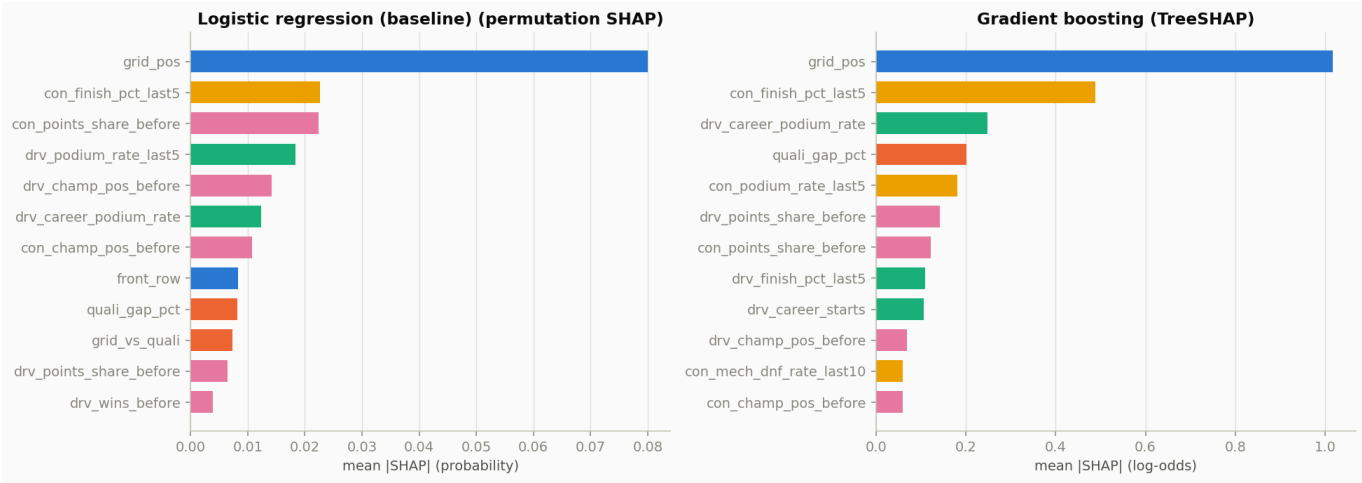
    - The order is `grid_pos`, then car strength (`con_finish_pct_last5`, `con_points_share_before`), then driver form and standings; the qualifying features come lower because the grid already carries most of their information.
    - All directions match F1 intuition. A grid penalty (`grid_vs_quali > 0`) *raises* the prediction for a given slot, because the driver is faster than the slot suggests.
    - TreeSHAP for gradient boosting also ranks `grid_pos` first by far.
- **Local** (2024 São Paulo GP, a wet race).
    - **Norris, pole: P = 0.78**, from a base of 0.156. Pole adds +0.29 (+0.04 more for the front row), team strength +0.19 and recent form +0.06. He finished 6th, a confident miss.
    - **Verstappen, 17th after a 5-place penalty: P = 0.05.** The grid removes −0.17 and a slow wet qualifying lap another −0.03, while the penalty (+0.04), his record and his team (about +0.02 each) partly compensate. He won.
    - LIME agrees qualitatively; its local fit is loose ($R^2 \approx 0.3$).



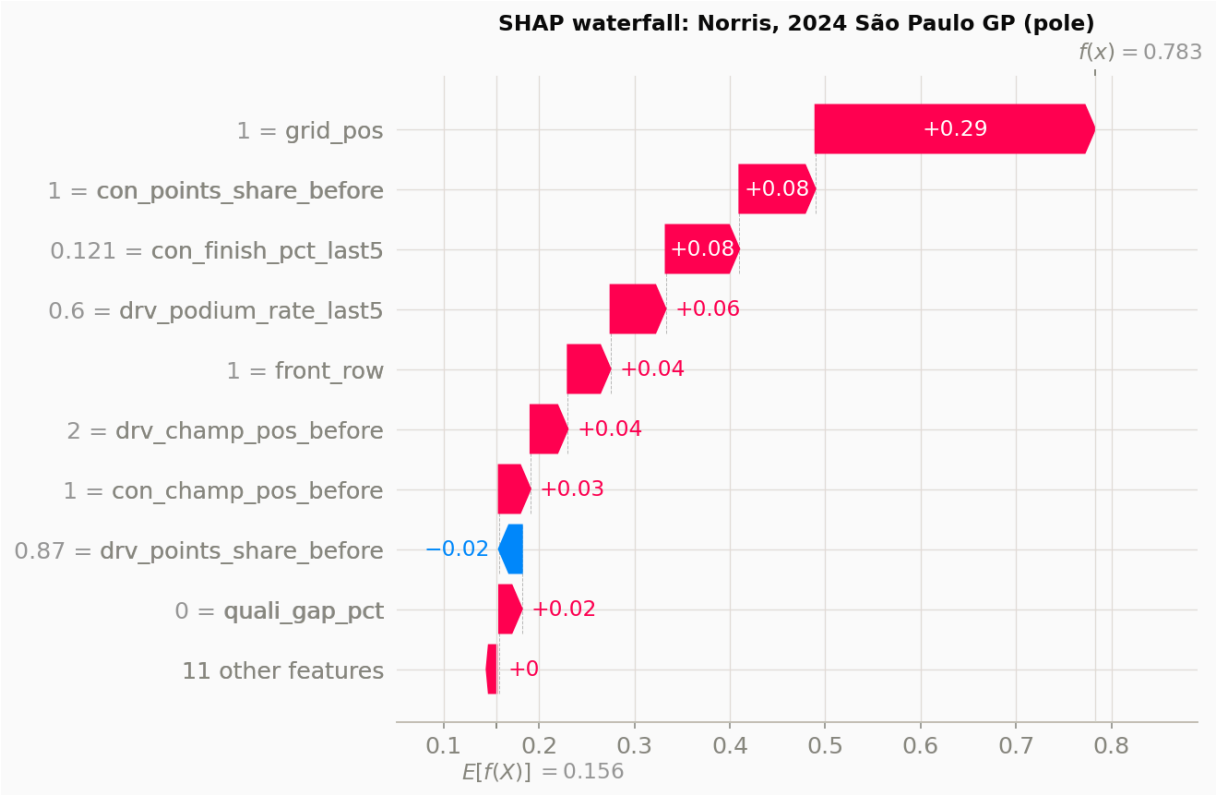
Global: permutation importance per feature and per feature group (validation PR-AUC drop).



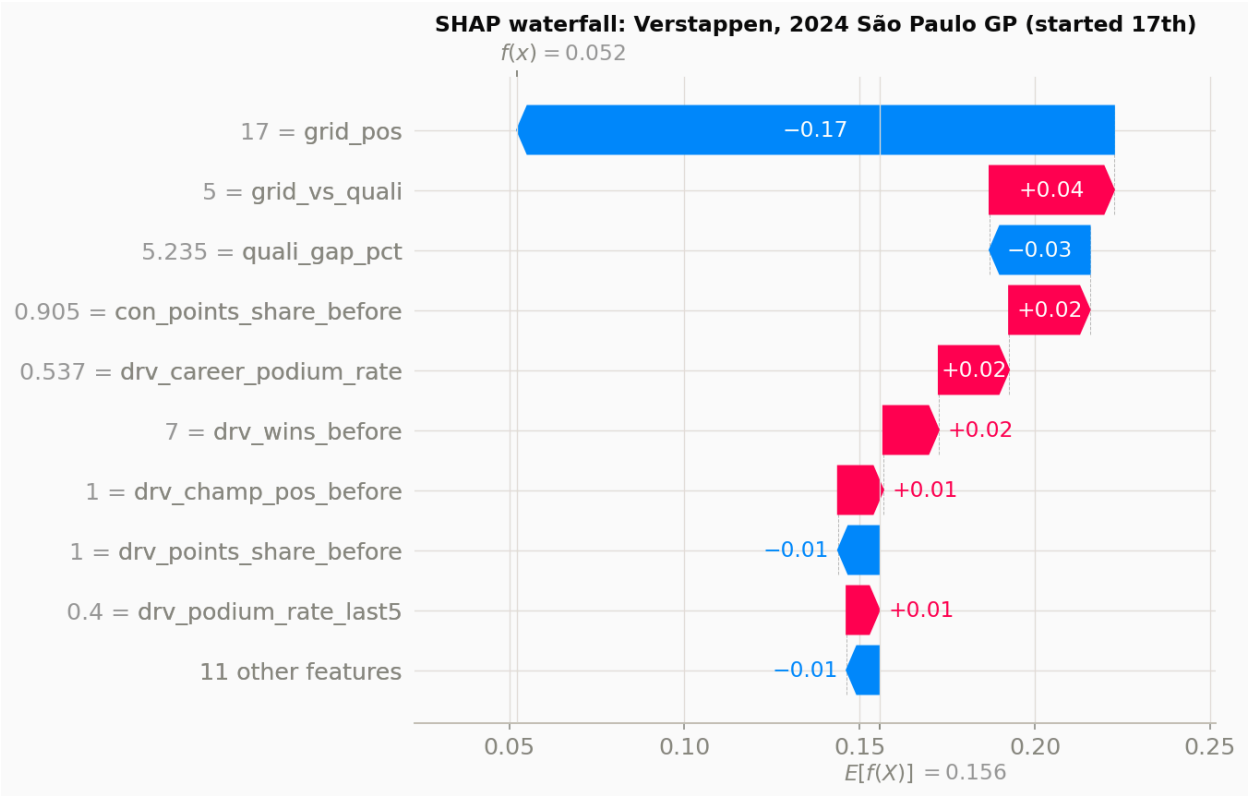
Global: SHAP summary on 300 test rows for the final model. Blue = low feature value, red = high.



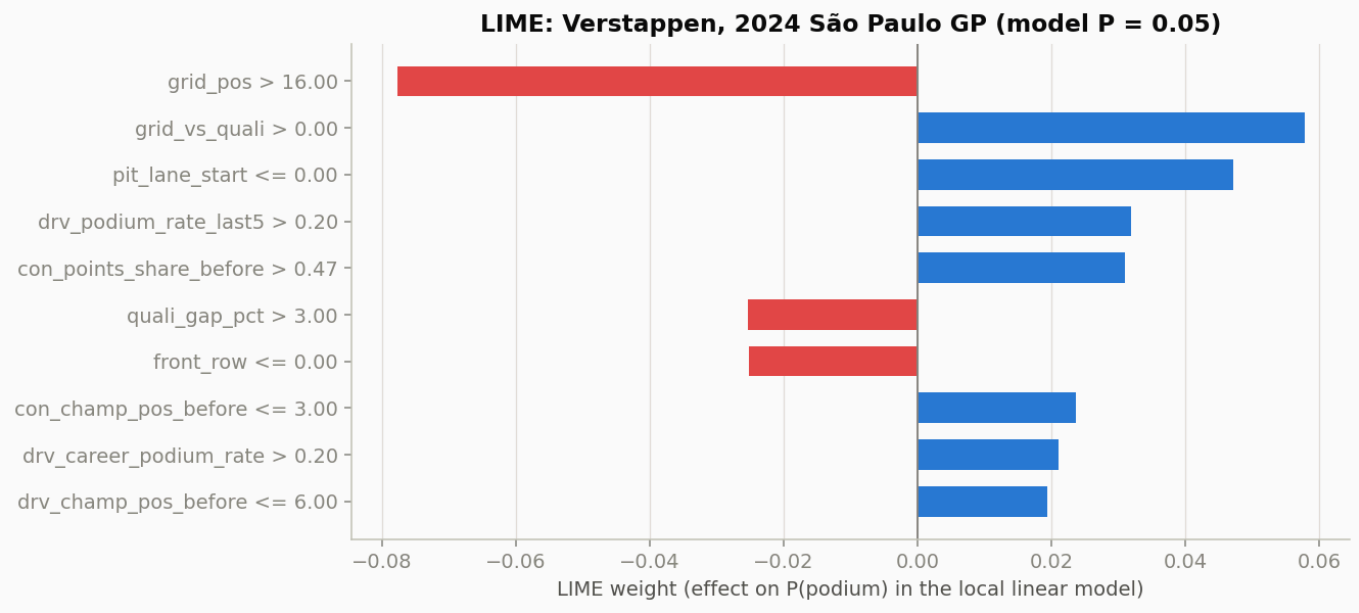
Global cross-check: mean |SHAP| for the final model vs gradient boosting (TreeSHAP). Both rank grid_pos first by a wide margin.



Local SHAP: Norris, 2024 São Paulo GP (pole, P = 0.78, finished 6th).



Local SHAP: Verstappen, 2024 São Paulo GP (started 17th, P = 0.05, won).



Local LIME: Verstappen, 2024 São Paulo GP. Red conditions push P(podium) down, blue push it up. Starting beyond P16 is the strongest negative, with a slow qualifying gap and no front-row start; his penalty (grid_vs_quali > 0), recent form and a front-running team push it up.

9. Inference function

`predict_podium(record)` takes a dictionary with any subset of the 20 features. It derives `front_row`, `grid_vs_quali` and `pit_lane_start` when possible, validates the inputs, imputes the missing ones inside the pipeline, and returns the probability, the decision at the validation threshold, and the list of missing inputs.

- **Complete record:** Verstappen, 2023 Japanese GP → P = 0.89, podium ✓.
- **Same record with its qualifying data removed:** P = 0.87, so the prediction degrades gracefully.
- **Very sparse records** are read as “debutant in round 1”, because that is what missing form and standings meant in training. The function reports `n_missing_inputs` as a warning.

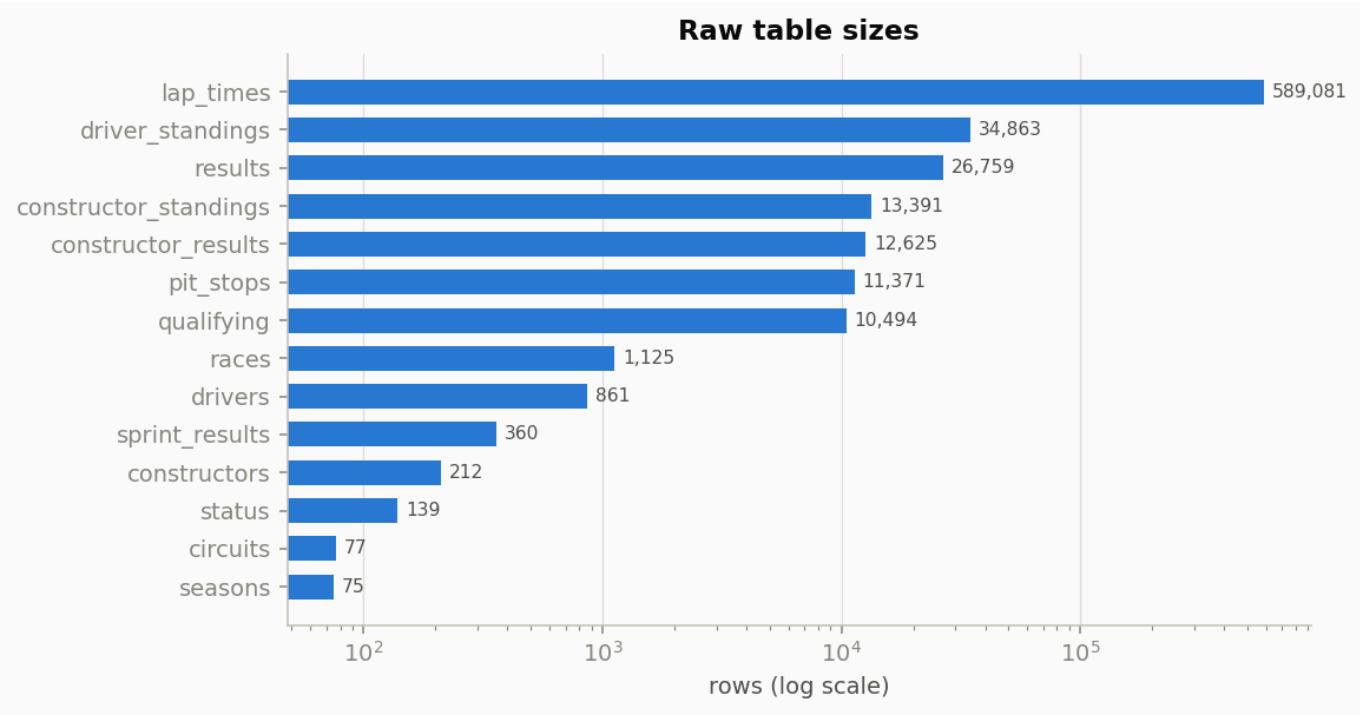
10. Limitations

1. **Qualifying-coverage drift.** No qualifying data before 1994, patchy data in 1996–2002, complete only from 2003. The 1994–2019 training data mixes seasons with and without qualifying records.
2. **Era sensitivity.** Reliability, field sizes, points systems and overtaking all changed; the grid-to-podium curve steepened over time. The 2022 rules are a distribution shift. The choice of training window is itself within noise (1950, 1981 and 1994 starts score within 0.002), and different library versions can tip it.
3. **Small evaluation sets** (117 and 204 podiums). The model differences are within noise, and selecting among many configurations inflates validation scores; the network’s lucky seed (0.739 vs a 0.727 mean) is an example.
4. **Race-day events** (rain, safety cars, crashes, failures) are invisible before the start.
5. **Constructor renames** reset team history.
6. **Documented judgement calls:** the shared-drive rule, the population filters, pit-lane placement and the status grouping.
7. **The explanations are not causal.**

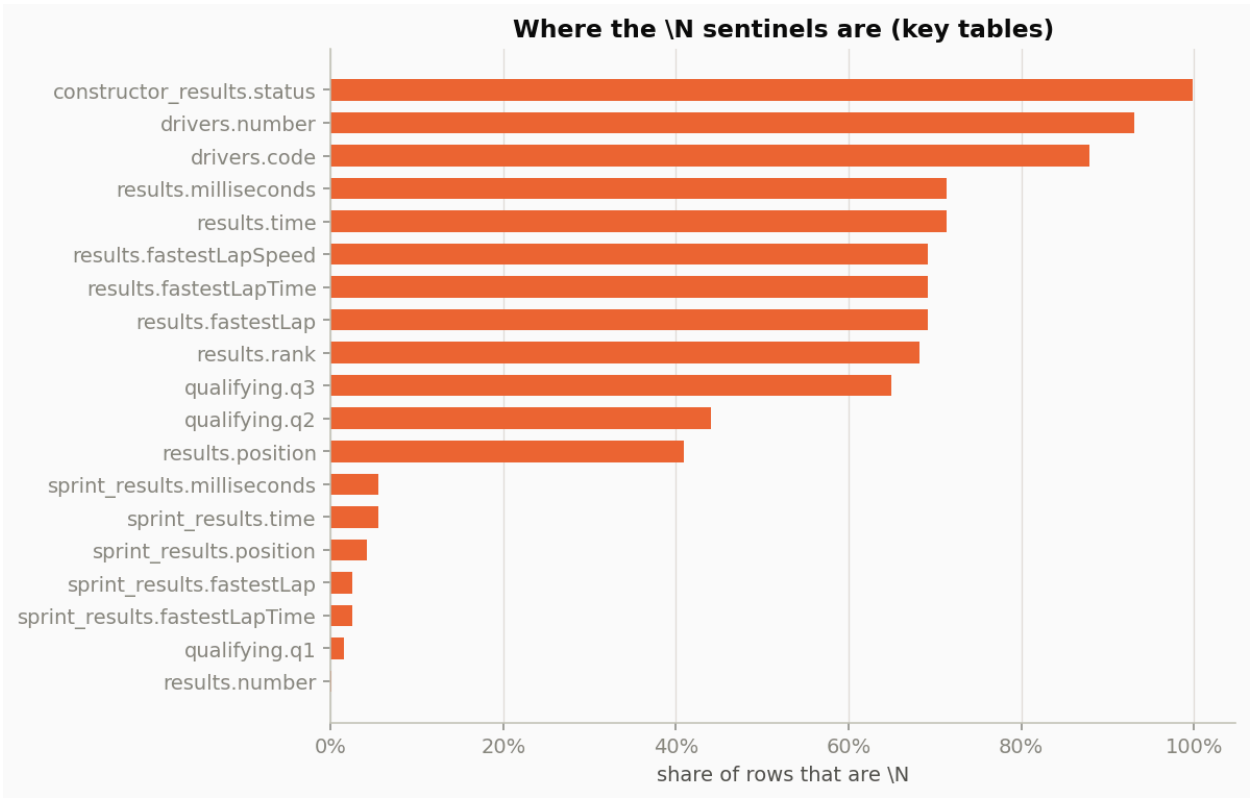
Appendix A. Additional figures

Every remaining figure from the notebook, in notebook order. The section numbers refer to the notebook.

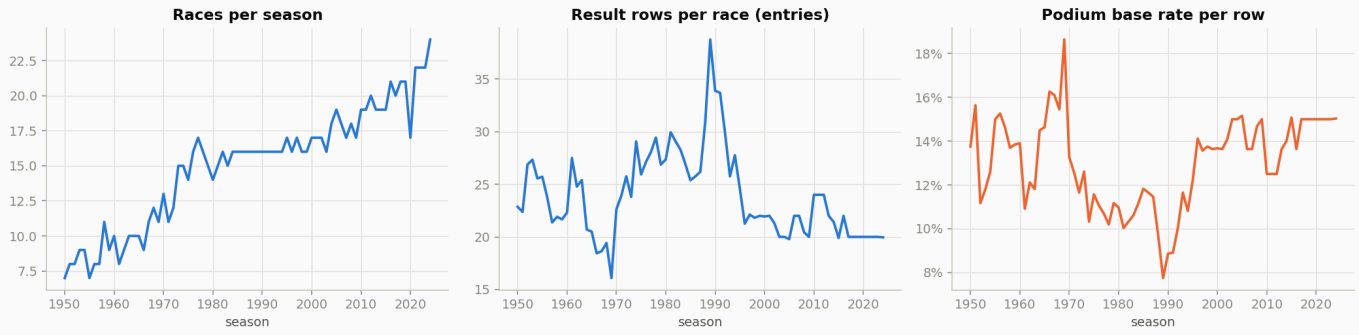
A.1 EDA before cleaning (notebook section 2)



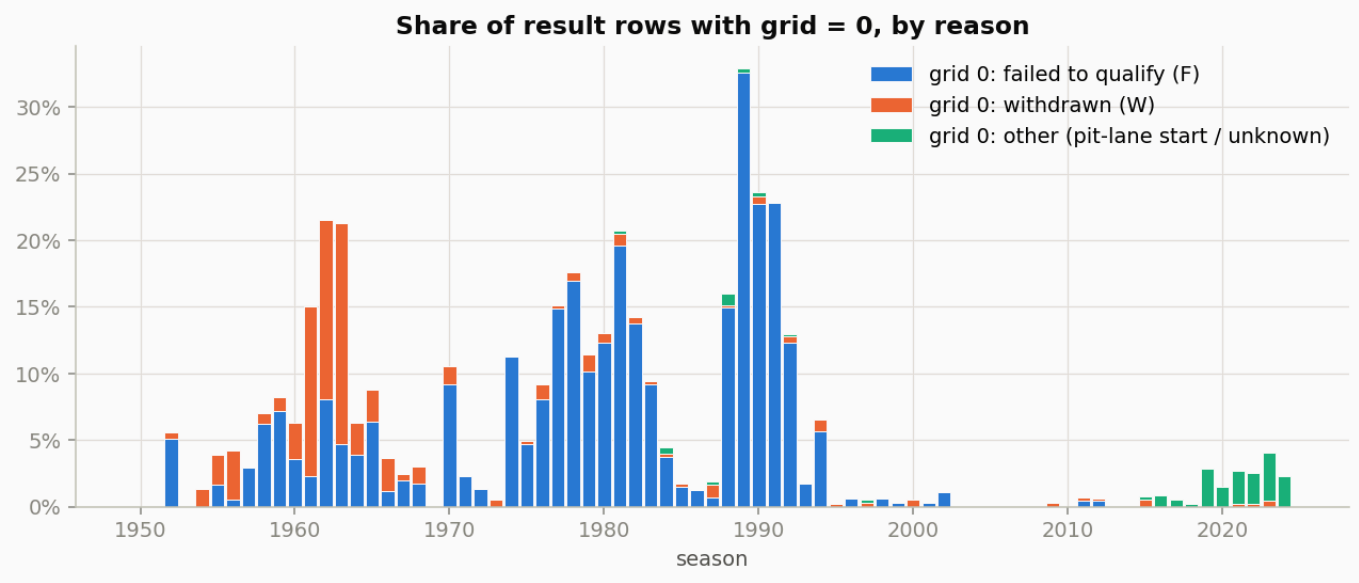
Raw table sizes (log scale). `lap_times` and `pit_stops` are post-race and never used as features.



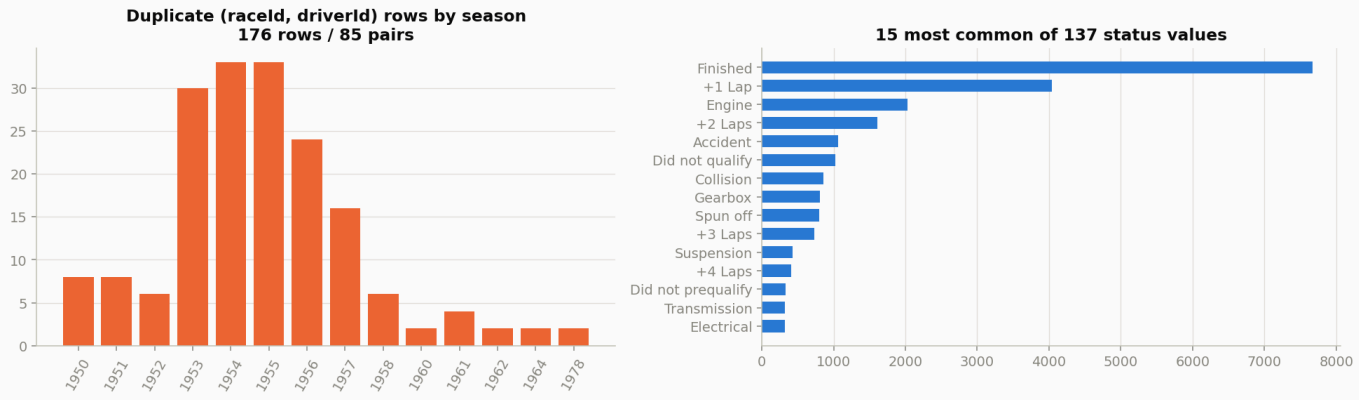
Share of rows that are `\N`, per column (key tables).



Raw results: races per season, entries per race and podium base rate per row.

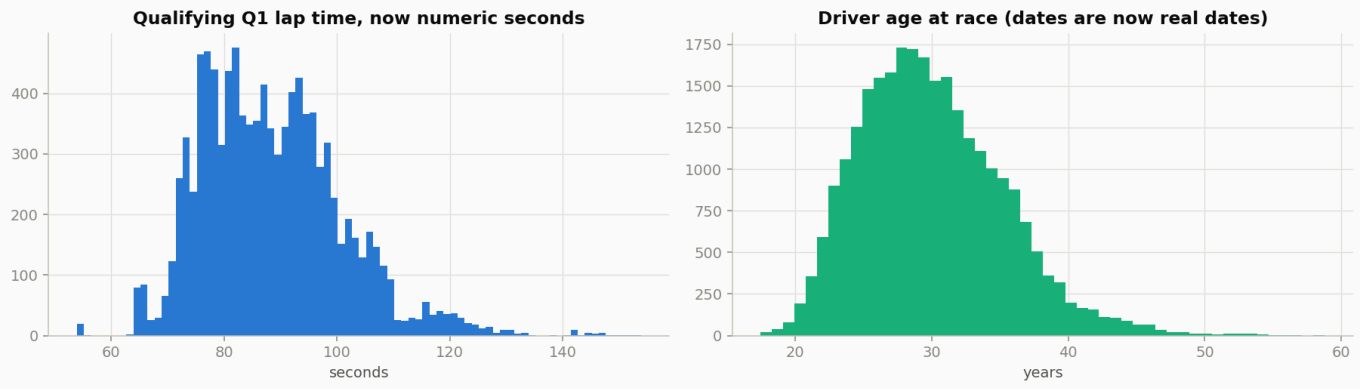


Rows with grid = 0 by reason: before 1994 almost always non-starters (F/W); from 1994 mostly pit-lane starts.

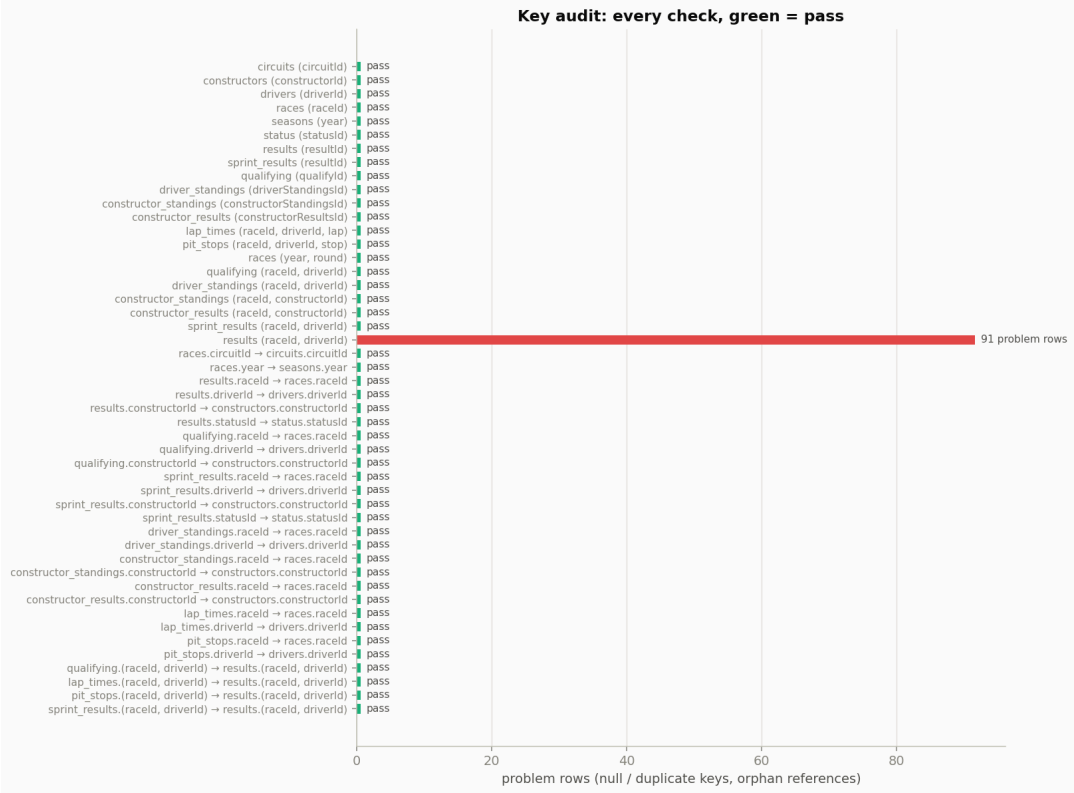


Duplicate (raceId, driverId) rows by season (shared drives) and the 15 most common status values.

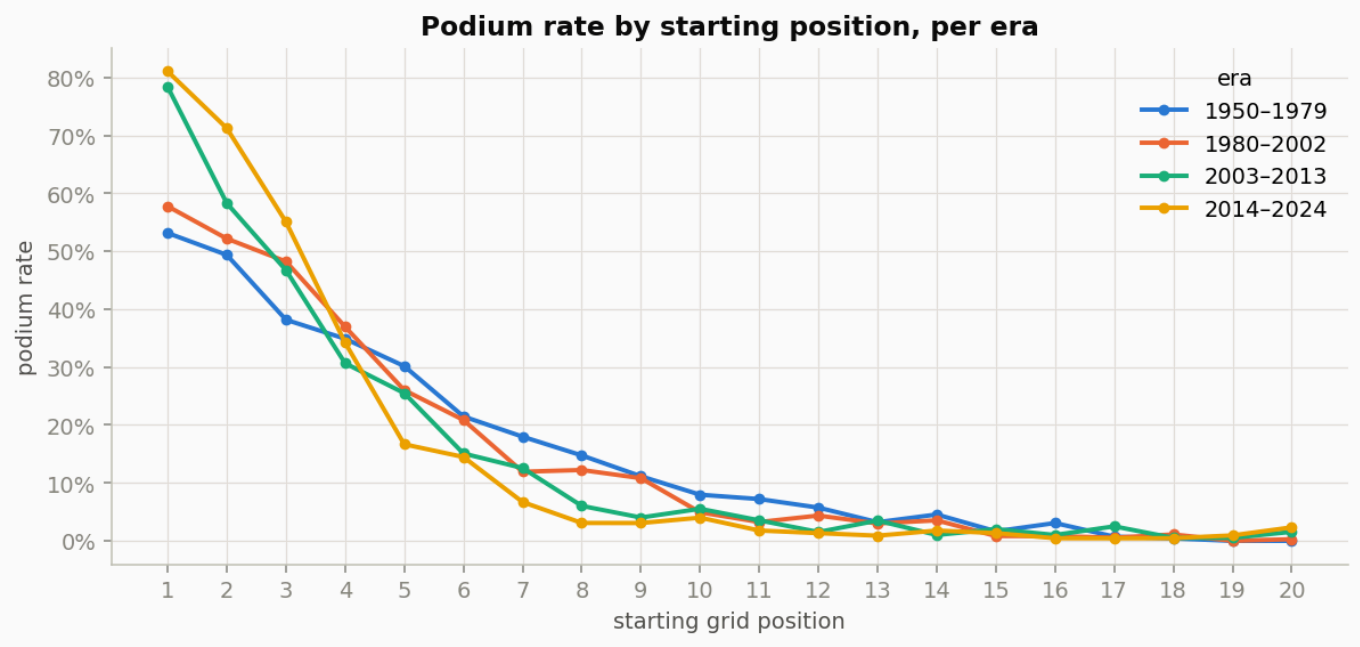
A.2 Cleaning and EDA after cleaning (notebook sections 3–4)



Semantic typing: qualifying Q1 times as numeric seconds and driver age at race from real dates.

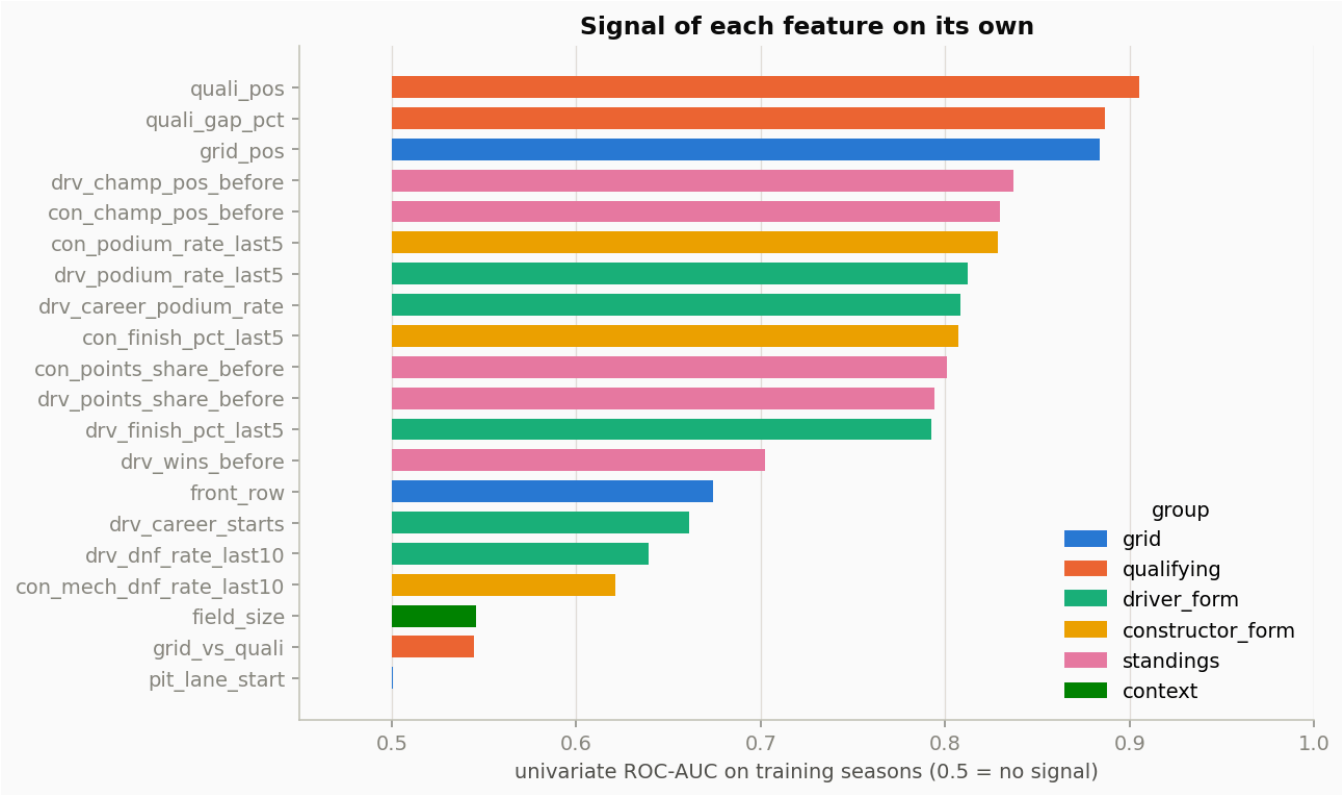


Key audit: every primary-key, natural-key and foreign-key check. Only the results grain (raceId, driverId) fails, before the duplicate rule is applied.

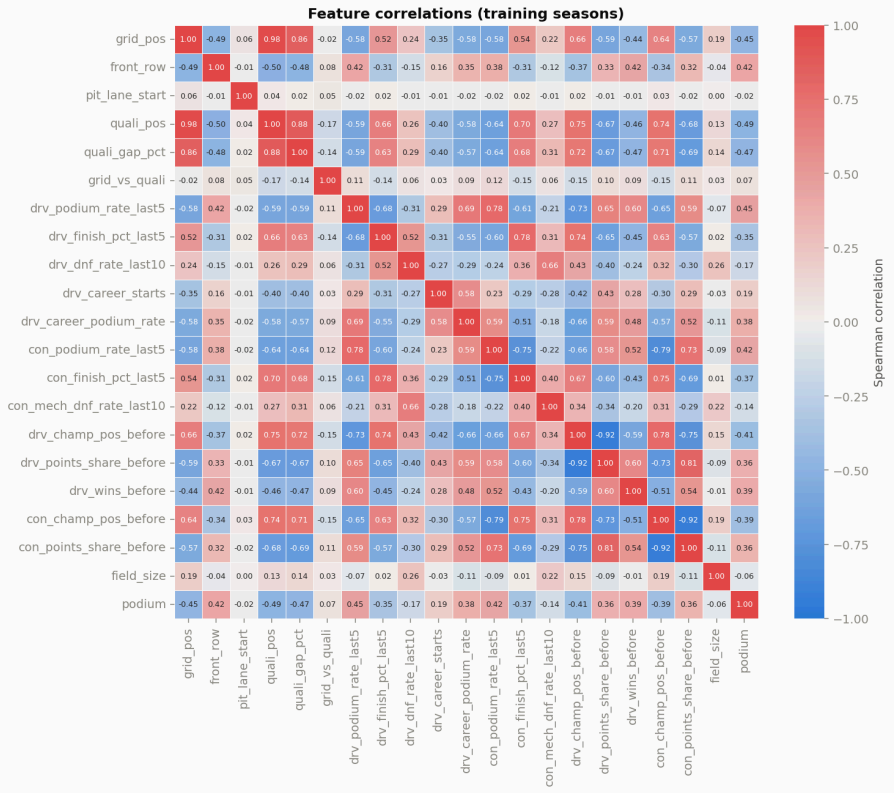


Podium rate by starting position per era: the curve steepens over time.

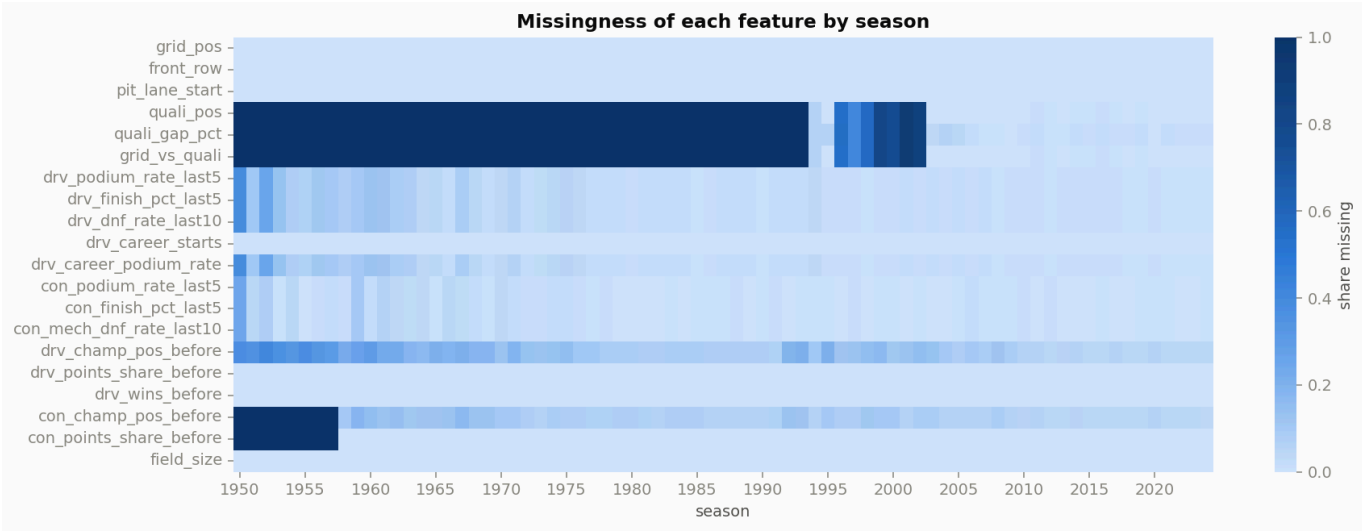
A.3 Feature evidence (notebook section 6)



Univariate ROC-AUC of each feature on the training seasons, coloured by feature group.

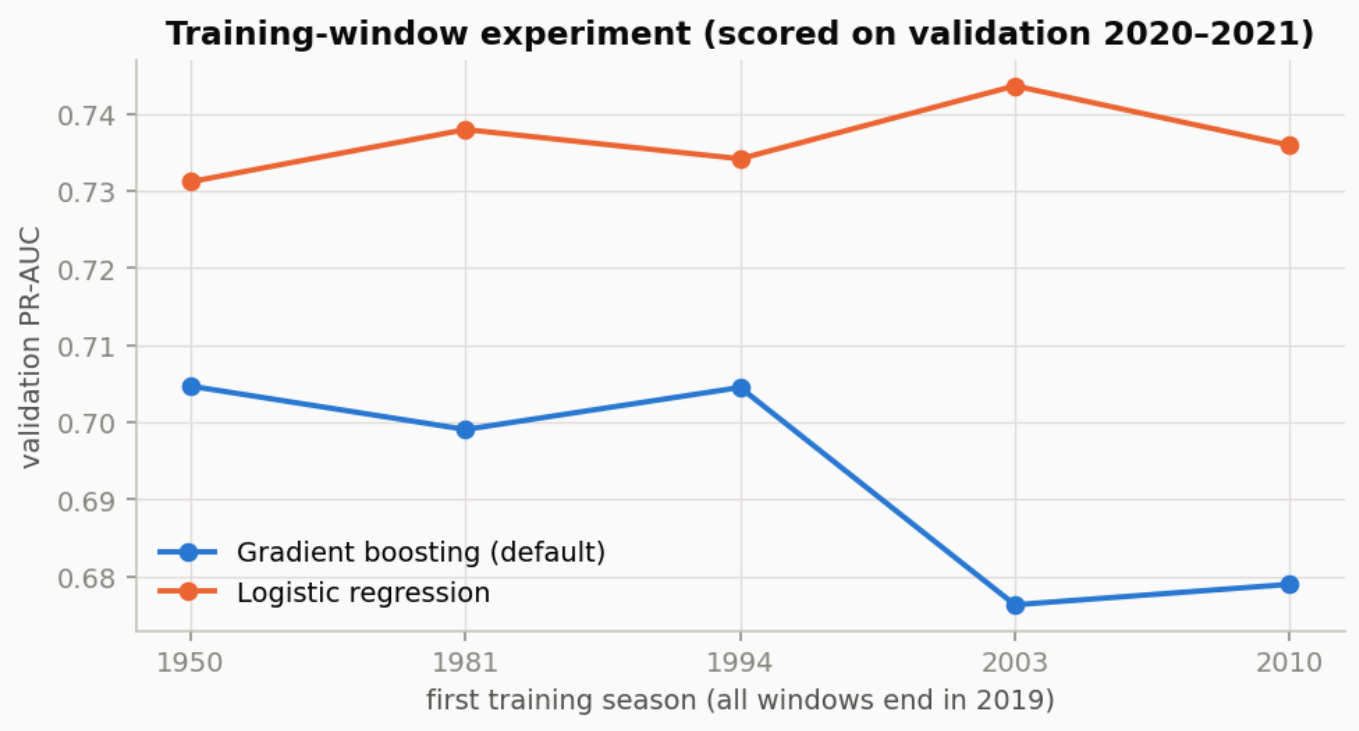


Spearman correlations between features and the target (training seasons).

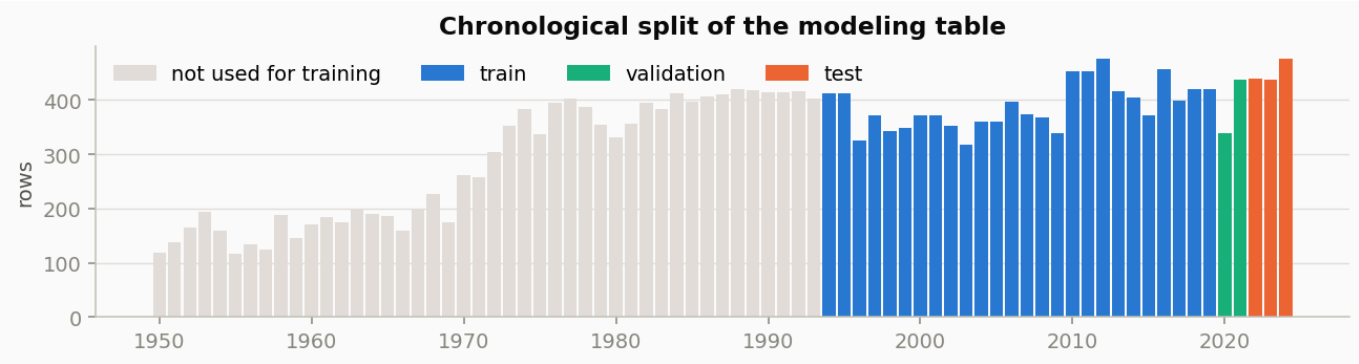


Missingness of each feature by season.

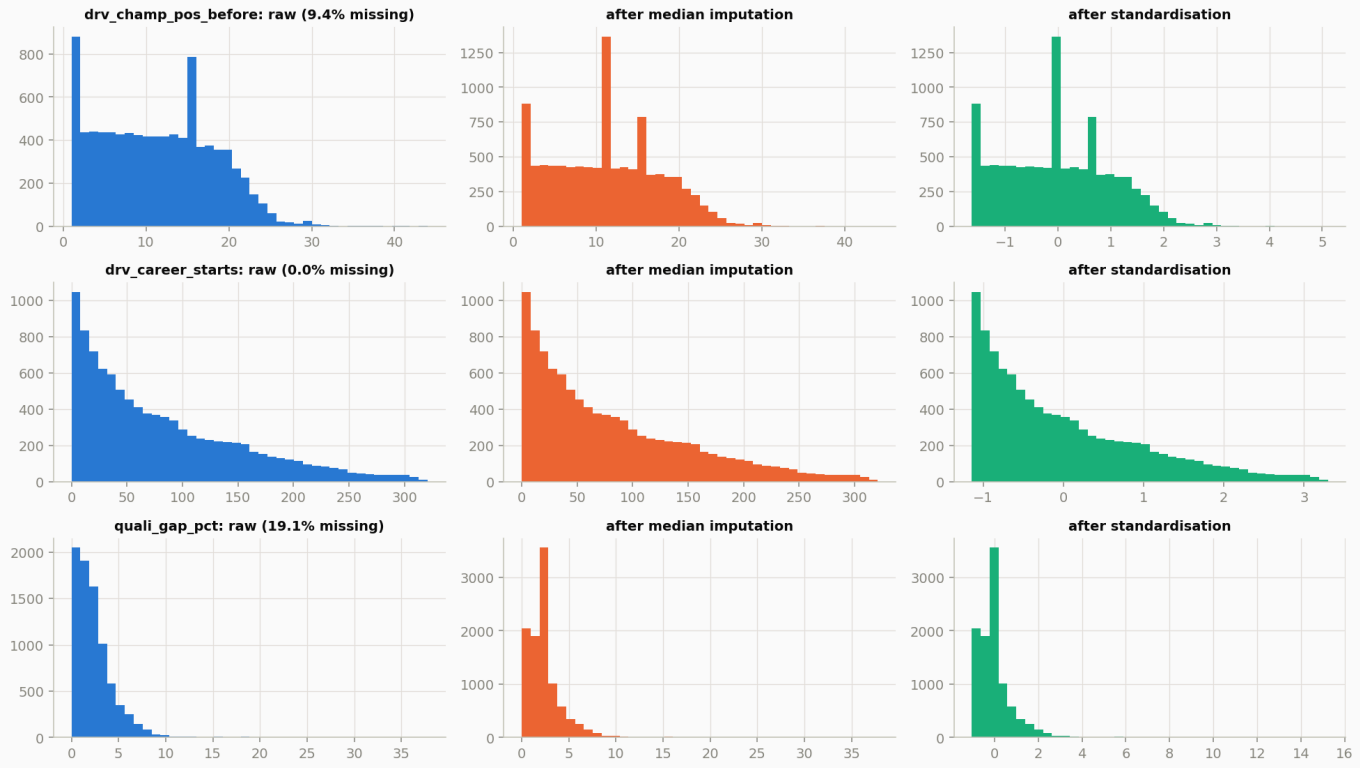
A.4 Split and preprocessing (notebook sections 7–8)



Training-window experiment: validation PR-AUC by first training season.

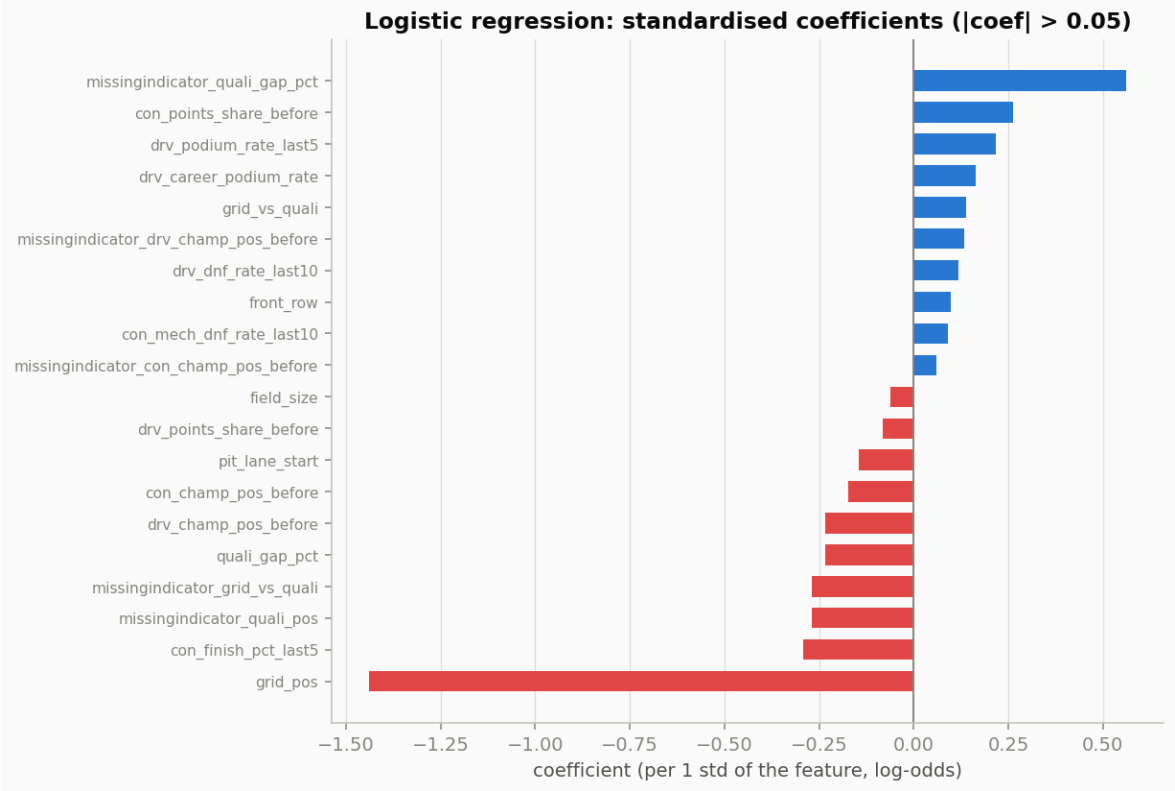


The chronological split of the modeling table.

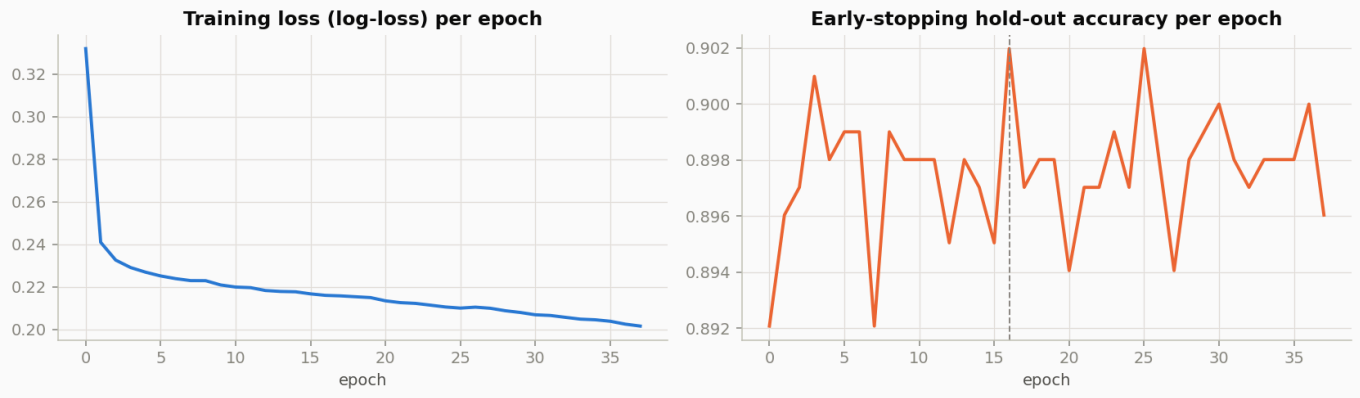


Effect of preprocessing on three features: raw → after median imputation → after standardisation.

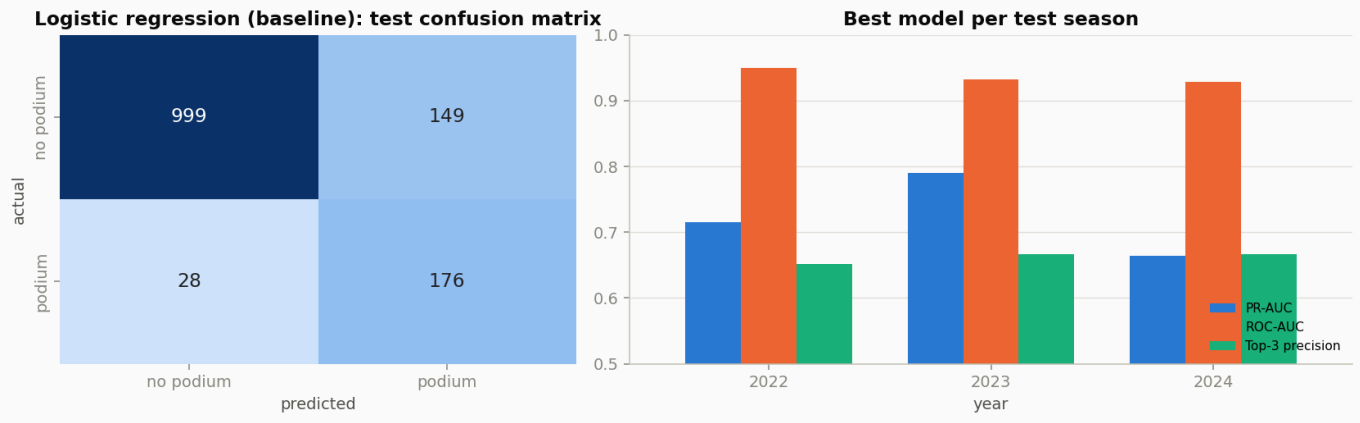
A.5 Models (notebook sections 9–10)



Logistic regression: standardised coefficients (|coef| > 0.05).



Shallow neural network: training loss and early-stopping hold-out score per epoch.



Final model: test confusion matrix at the validation threshold, and scores per test season.